Maps from a truncation into an n-type correspond to maps from the original type into it

The Univalent Foundations Program

hott-book:hlevels.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author lem environment at source lines 541-546 and its proof environment opens on line 548, the line after the blank separator; the pairing is by adjacency and the retained proof range 548-552 is the whole environment.

Editorial note. The book label-number dump gives this label as 7.3.3 (env lem, chapter 7, section 3, third counter value), so the counters chapter 7, section 3 and thm 2 (zero-based) are set so that the item prints with the book number.

Editorial note. The counted statement and proof refer to no label outside the card; they use the recursion principle of truncation and function extensionality, both recorded as standard prerequisites rather than treated as delegations.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: the universal property of truncations (author label thm:trunc-reflective, book Lemma 7.3.3).

Allowed standard prerequisites (editorial).

- The recursion principle of truncation. A map from a type into an n-truncated type extends to the truncation of its source, and the extension sends a projection to the value of the original map.. It supplies both the extension used in one direction and the equality checks in the proof.
- Function extensionality. Two maps are equal as soon as they agree on every argument.. The two round-trip verifications are carried out pointwise on generators.

Lemma 7.3.3 (Universal property of truncations)**.** *Let $n \geq -2$, $A : \mathcal{U}$ and $B : n\text{-}\mathsf{Type}$. The following map is an equivalence:*

$$\begin{cases} (\|A\|_n \to B) & \longrightarrow & (A \to B) \\ g & \longmapsto & g \circ |-|_n \end{cases}$$

Proof. Given that B is n-truncated, any $f : A \to B$ can be extended to a map $\mathsf{ext}(f) : \|A\|_n \to B$. The map $\mathsf{ext}(f) \circ |-|_n$ is equal to f, because for every $a : A$ we have $\mathsf{ext}(f)(|a|_n) = f(a)$ by definition. And the map $\mathsf{ext}(g \circ |-|_n)$ is equal to g, because they both send $|a|_n$ to $g(|a|_n)$. □